数学物理方法 · 第 15 讲
Mathematical Methods for Physics
Lecture 15

卷积、傅里叶变换的应用与拉普拉斯变换

Convolution, Applications of Fourier Transforms, and the Laplace Transform

一、卷积与傅里叶变换
Convolution and Fourier Transforms

1. 卷积的定义及运算性质
Definition and Algebraic Properties of Convolution

接第 *14* 讲的 *"性质 8: 卷积性"*。

continuation of "Property 8: Convolution" in Lecture 14.

定义（卷积）

Definition (convolution)

定义在 $(-\infty,+\infty)$ 上的两个函数 $f_1(t)$、$f_2(t)$，其卷积定义为

For two functions $f_1(t)$ and $f_2(t)$ defined on $(-\infty,+\infty)$, their convolution is defined by

$$(f_1 * f_2)(t) = \int_{-\infty}^{+\infty} f_1(\tau) f_2(t-\tau)\,\mathrm{d}\tau.$$

运算性质

Algebraic properties

$$f_1 * f_2 = f_2 * f_1,$$

$$f_1 * (f_2 * f_3) = (f_1 * f_2) * f_3,$$

$$f_1 * (f_2 + f_3) = f_1 * f_2 + f_1 * f_3.$$

Notes:

2. 卷积定理
The Convolution Theorem

设

Let

$$\mathcal{F}[f_1(t)] = F_1(\omega), \qquad \mathcal{F}[f_2(t)] = F_2(\omega).$$

（1）卷积的傅里叶变换等于变换的乘积

(1) The Fourier transform of a convolution is the product of the transforms

$$\mathcal{F}[(f_1 * f_2)(t)] = \mathcal{F}[f_1] \cdot \mathcal{F}[f_2] = F_1(\omega)F_2(\omega).$$

即

That is,

$$f_1 * f_2 \overset{\mathcal{F}}{\leftrightarrow} F_1F_2.$$

（2）乘积的傅里叶变换

(2) Fourier transform of a product

$$\mathcal{F}[f_1f_2] = \frac{1}{2\pi}\big(\mathcal{F}[f_1] * \mathcal{F}[f_2]\big) = \frac{1}{2\pi}(F_1 * F_2).$$

证明（1）

Proof of (1)

$$\begin{aligned}\mathcal{F}[(f_1 * f_2)(t)] &= \int_{-\infty}^{+\infty} (f_1 * f_2)(t)e^{-i\omega t}\,\mathrm{d}t \\ &= \int_{-\infty}^{+\infty} \left[\int_{-\infty}^{+\infty} f_1(\tau)f_2(t-\tau)\,\mathrm{d}\tau\right] e^{-i\omega t}\,\mathrm{d}t \\ &= \int_{-\infty}^{+\infty} f_1(\tau)\left[\int_{-\infty}^{+\infty} f_2(t-\tau)e^{-i\omega t}\,\mathrm{d}t\right]\mathrm{d}\tau.\end{aligned}$$

利用平移性，

By the shifting property,

$$\int_{-\infty}^{+\infty} f_2(t-\tau)e^{-i\omega t}\,\mathrm{d}t = e^{-i\tau\omega}F_2(\omega).$$

因此

Therefore,

$$\begin{aligned}\mathcal{F}[(f_1 * f_2)(t)] &= \int_{-\infty}^{+\infty} f_1(\tau)e^{-i\tau\omega}F_2(\omega)\,\mathrm{d}\tau \\ &= F_2(\omega)\int_{-\infty}^{+\infty} f_1(\tau)e^{-i\tau\omega}\,\mathrm{d}\tau \\ &= F_2(\omega)F_1(\omega).\end{aligned}$$

Notes:

3. 例：用卷积证明积分性质

Example: Proving the Integration Property by Convolution

若

If

$$\int_{-\infty}^{+\infty} f(t)\,\mathrm{d}t \neq 0,$$

则

then

$$\mathcal{F}\left[\int_{-\infty}^{t} f(\tau)\,\mathrm{d}\tau\right] = \frac{F(\omega)}{i\omega} + \pi\delta(\omega)F(0).$$

证明

Proof

先说明

First, we show that

$$\int_{-\infty}^{t} f(\tau)\,\mathrm{d}\tau = (f * u)(t),$$

其中

where

$$u(t) = \begin{cases} 1, & t > 0, \\ 0, & t < 0. \end{cases}$$

事实上，

Indeed,

$$(f * u)(t) = \int_{-\infty}^{+\infty} f(\tau)u(t-\tau)\,\mathrm{d}\tau.$$

令 $t - \tau = \xi$，则

Let $t - \tau = \xi$. Then

$$\begin{aligned}(f * u)(t) &= \int_{+\infty}^{-\infty} f(t-\xi)u(\xi)(-\mathrm{d}\xi) \\ &= \int_{-\infty}^{+\infty} f(t-\xi)u(\xi)\,\mathrm{d}\xi \\ &= \int_{-\infty}^{0} 0\,\mathrm{d}\xi + \int_{0}^{+\infty} f(t-\xi)\,\mathrm{d}\xi.\end{aligned}$$

再令 $t - \xi = \tau$，得

Now let $t - \xi = \tau$. We obtain

$$(f * u)(t) = \int_{t}^{-\infty} f(\tau)(-\mathrm{d}\tau) = \int_{-\infty}^{t} f(\tau)\,\mathrm{d}\tau.$$

由卷积定理，
By the convolution theorem,

$$\begin{aligned}\mathcal{F}\left[\int_{-\infty}^{t} f(\tau)\,\mathrm{d}\tau\right] &= \mathcal{F}[(f*u)(t)] \\ &= \mathcal{F}[f]\cdot\mathcal{F}[u] \\ &= F(\omega)\left[\frac{1}{i\omega}+\pi\delta(\omega)\right] \\ &= \frac{F(\omega)}{i\omega}+F(\omega)\pi\delta(\omega) \\ &= \frac{F(\omega)}{i\omega}+F(0)\pi\delta(\omega).\end{aligned}$$

Notes:

二、傅里叶变换的应用
Applications of Fourier Transforms

教材 *P.200*。
textbook, p. 200.

原稿列出两类应用：
The original notes list two types of application.

- 求解微分方程。
  Solving differential equations.
- 求解积分方程。
  Solving integral equations.

1. 例 7.20：求解微分方程
Example 7.20: Solving a Differential Equation

求解
Solve

$$x''(t)-x(t)=-f(t),\qquad -\infty<t<+\infty.$$

原稿蓝字补充：为使用微分性，假设
An annotation in blue in the original introduces the following assumptions for applying the differentiation property.

$$x(\pm\infty)=0,\qquad x'(\pm\infty)=0.$$

解
Solution

第一步，取傅里叶变换。

Step 1. Take the Fourier transform.

设

Let

$$\mathcal{F}[x(t)] = \overline{X}(\omega), \qquad \mathcal{F}[f(t)] = F(\omega).$$

对方程两边取变换，

Taking the Fourier transform of both sides gives

$$\mathcal{F}[x''(t) - x(t)] = -\mathcal{F}[f(t)].$$

由微分性，

By the differentiation property,

$$(i\omega)^2\overline{X}(\omega) - \overline{X}(\omega) = -F(\omega),$$

所以

Therefore,

$$\overline{X}(\omega) = \frac{F(\omega)}{\omega^2 + 1}.$$

第二步，取傅里叶反变换。

Step 2. Take the inverse Fourier transform.

$$\begin{aligned} x(t) &= \mathcal{F}^{-1}[\overline{X}(\omega)] \\ &= \mathcal{F}^{-1}\left[F(\omega)\frac{1}{\omega^2+1}\right] \\ &= \mathcal{F}^{-1}[F(\omega)] * \mathcal{F}^{-1}\left[\frac{1}{\omega^2+1}\right] \\ &= f(t) * \left(\frac{1}{2}e^{-|t|}\right) \\ &= \frac{1}{2}\int_{-\infty}^{+\infty} f(\tau)e^{-|t-\tau|}\,\mathrm{d}\tau. \end{aligned}$$

这就是原稿所求的解。

This is the solution obtained in the original notes.

其中使用了变换对

The transform pair used here is

$$e^{-\beta|t|} \overset{\mathcal{F}}{\longleftrightarrow} \frac{2\beta}{\omega^2 + \beta^2},$$

从而

It follows that

$$\mathcal{F}^{-1}\left[\frac{1}{\omega^2+\beta^2}\right] = \frac{1}{2\beta}e^{-\beta|t|}, \qquad \beta > 0.$$

整理注：$\beta > 0$ 为此衰减指数变换对的条件。
Editorial note. The condition $\beta > 0$ is required for this decaying-exponential transform pair.

Notes:

2. 例 7.22：求解卷积型积分方程

Example 7.22: Solving a Convolution Integral Equation

求解积分方程
Solve the integral equation

$$\int_{-\infty}^{+\infty} e^{-|t-\xi|} x(\xi)\,\mathrm{d}\xi = f(t),$$

其中 $f(t)$ 已知。
where $f(t)$ is known.

分析：左端是一个卷积，原方程可写为
Analysis. The left-hand side is a convolution, so the equation can be written as

$$e^{-|t|} * x(t) = f(t).$$

第一步，取傅里叶变换。
Step 1. Take the Fourier transform.

利用卷积定理，
By the convolution theorem,

$$\mathcal{F}[e^{-|t|} * x(t)] = \mathcal{F}[f(t)],$$

$$\mathcal{F}[e^{-|t|}] \cdot \mathcal{F}[x(t)] = F(\omega).$$

即
That is,

$$\frac{2}{\omega^2+1}\overline{X}(\omega) = F(\omega).$$

所以
Therefore,

$$\overline{X}(\omega) = \frac{\omega^2+1}{2}F(\omega).$$

第二步，取傅里叶反变换。
Step 2. Take the inverse Fourier transform.

$$\begin{aligned} x(t) &= \mathcal{F}^{-1}\left[\frac{\omega^2+1}{2}F(\omega)\right] \\ &= \frac{1}{2}\left\{\mathcal{F}^{-1}[\omega^2 F(\omega)] + \mathcal{F}^{-1}[F(\omega)]\right\} \\ &= \frac{1}{2}\left\{-\mathcal{F}^{-1}[(i\omega)^2 F(\omega)] + f(t)\right\} \\ &= \frac{1}{2}[-f''(t) + f(t)]. \end{aligned}$$

注：微分性
Note: the differentiation property

$$f''(t) \overset{\mathcal{F}}{\leftrightarrow} (i\omega)^2 F(\omega),$$

条件为
The conditions are

$$f(\pm\infty) = 0, \qquad f'(\pm\infty) = 0.$$

特别地，设
In particular, let

$$f(t) = e^{-|t|} - \frac{1}{2}e^{-2|t|},$$

则可以代入上述公式求解。
The solution can then be obtained by substitution into the formula above.

Notes:

3. 例 7.21：求解正弦核积分方程
Example 7.21: Solving an Integral Equation with a Sine Kernel

求解
Solve

$$\int_0^{+\infty} x(\omega)\sin\omega t\,\mathrm{d}\omega = e^{-t}, \qquad t > 0.$$

回忆：傅里叶积分公式（反变换）

Review: The Fourier Integral Formula (Inverse Transform)

$$f(t)=\frac{1}{2\pi}\int_{-\infty}^{+\infty}\left[\int_{-\infty}^{+\infty}f(\tau)e^{-i\omega\tau}\,\mathrm{d}\tau\right]e^{i\omega t}\,\mathrm{d}\omega.$$

特别地

Special cases

(I) 偶函数

(I) Even functions

若 $f(-t)=f(t)$，则

If $f(-t)=f(t)$, then

$$f(t)=\frac{2}{\pi}\int_{0}^{+\infty}\left[\int_{0}^{+\infty}f(\tau)\cos\omega\tau\,\mathrm{d}\tau\right]\cos\omega t\,\mathrm{d}\omega.$$

称为**余弦傅里叶积分公式**。

This is called the **Fourier cosine integral formula**.

(II) 奇函数

(II) Odd functions

若 $f(-t)=-f(t)$，则

If $f(-t)=-f(t)$, then

$$f(t)=\frac{2}{\pi}\int_{0}^{+\infty}\left[\int_{0}^{+\infty}f(\tau)\sin\omega\tau\,\mathrm{d}\tau\right]\sin\omega t\,\mathrm{d}\omega.$$

称为**正弦傅里叶积分公式**。

This is called the **Fourier sine integral formula**.

回到原题

Return to the Integral Equation

由

From

$$\int_{0}^{+\infty}x(\omega)\sin\omega t\,\mathrm{d}\omega=e^{-t},$$

得

we obtain

$$\frac{2}{\pi}\int_{0}^{+\infty}x(\omega)\sin\omega t\,\mathrm{d}\omega=\frac{2}{\pi}e^{-t}.$$

由正弦积分公式可知

The Fourier sine integral formula gives

$$x(\omega)=\int_{0}^{+\infty}\frac{2}{\pi}e^{-t}\sin\omega t\,\mathrm{d}t.$$

列出两种计算方法：
Here suggests two methods of evaluating the integral.

- 方法一：两次分部积分。
  Method 1: integrate by parts twice.
- 方法二：利用 Euler 公式
  Method 2: use Euler's formula,

$$\sin \omega t = \frac{e^{i\omega t} - e^{-i\omega t}}{2i}.$$

Notes:

三、第 8 章：拉普拉斯变换
Chapter 8: The Laplace Transform

1. 拉普拉斯变换的定义
Definition of the Laplace Transform

拉普拉斯变换，简称拉氏变换。
In Chinese, the name "Laplace transform" is also abbreviated as "拉氏变换".

对于定义在 $[0, +\infty)$ 上的函数 $f(t)$，定义
For a function $f(t)$ defined on $[0, +\infty)$, define

$$F(p) = \int_0^{+\infty} f(t) e^{-pt} \, \mathrm{d}t,$$

其中
where

$$p = \sigma + i\omega$$

是复变量。
is a complex variable.

称 $\{f(t), F(p)\}$ 为一个拉氏变换对，记作
The pair $\{f(t), F(p)\}$ is called a Laplace transform pair. We write

$$\mathcal{L}[f(t)] = F(p),$$

或
or

$$\mathcal{L}^{-1}[F(p)] = f(t).$$

Notes:

2. 指数级函数与拉氏变换存在定理

Functions of Exponential Order and the Existence Theorem

定义：指数级函数

Definition: A Function of Exponential Order

若

If

$$|f(t)| \leq Me^{\sigma_c t},$$

则称 $f(t)$ 是“指数级函数”，σ_c 是其增长指数。$\sigma_c \geq 0$。

then $f(t)$ is said to be of exponential order, and σ_c is its growth exponent. $\sigma_c \geq 0$.

定理 8.1：拉氏变换存在定理

Theorem 8.1: Existence of the Laplace Transform

若定义在 $[0, +\infty)$ 上的函数 $f(t)$ 满足

Suppose that $f(t)$, defined on $[0, +\infty)$, satisfies the following conditions.

1. $f(t)$ 满足 Dirichlet 条件；
   $f(t)$ satisfies the Dirichlet conditions.
2. $f(t)$ 是指数级的，即 $|f(t)| \leq Me^{\sigma_c t}$；
   $f(t)$ is of exponential order, that is, $|f(t)| \leq Me^{\sigma_c t}$.

则

then

$$F(p) = \int_0^{+\infty} f(t)e^{-pt}\,\mathrm{d}t$$

在右半平面

in the right half-plane

$$\mathrm{Re}\, p > \sigma_c$$

内存在且解析。

exists and is analytic.

注：令 $\widetilde{f}(t) = f(t)u(t)$。

note: let $\widetilde{f}(t) = f(t)u(t)$.

原图说明：复平面以横轴表示 $\sigma = \operatorname{Re} p$，纵轴表示 $\omega = \operatorname{Im} p$；直线 $\sigma = \sigma_c$ 右侧的区域用斜线标出，表示收敛与解析的右半平面。

Description of the original figure. In the complex plane, the horizontal axis is $\sigma = \operatorname{Re} p$ and the vertical axis is $\omega = \operatorname{Im} p$. The region to the right of $\sigma = \sigma_c$ is shaded to indicate the right half-plane of convergence and analyticity.

证明：绝对收敛

Proof: Absolute Convergence

因为

Since

$$|f(t)| \le M e^{\sigma_c t}$$

且

and

$$|e^{-pt}| = |e^{-\sigma t} e^{-i\omega t}| = e^{-\sigma t},$$

所以当 $\sigma > \sigma_c$ 时，

for $\sigma > \sigma_c$ we have

$$\begin{aligned}
\int_0^{+\infty} |f(t) e^{-pt}| \, \mathrm{d}t &\le \int_0^{+\infty} M e^{\sigma_c t} |e^{-pt}| \, \mathrm{d}t \\
&= \int_0^{+\infty} M e^{-(\sigma - \sigma_c) t} \, \mathrm{d}t \\
&= M \frac{e^{-(\sigma - \sigma_c) t}}{-(\sigma - \sigma_c)} \Bigg|_0^{+\infty} \\
&= \frac{M}{\sigma - \sigma_c}.
\end{aligned}$$

因此广义积分

Therefore, the improper integral

$$\int_0^{+\infty} f(t) e^{-pt} \, \mathrm{d}t$$

存在，且绝对收敛。

exists and converges absolutely.

证明：解析性

Proof: Analyticity

原稿进一步说明原积分以及

The original notes further state that both the original integral and

$$\int_0^{+\infty} f(t) \frac{\partial}{\partial p} (e^{-pt}) \, \mathrm{d}t$$

具有一致收敛性，从而可对参数 p 求导：
converge uniformly, allowing differentiation with respect to the parameter p.

$$\begin{aligned} F'(p) &= \frac{\mathrm{d}}{\mathrm{d}p}\int_0^{+\infty} f(t)e^{-pt}\,\mathrm{d}t \\ &= \int_0^{+\infty} f(t)\frac{\partial}{\partial p}(e^{-pt})\,\mathrm{d}t \\ &= \int_0^{+\infty} (-t)f(t)e^{-pt}\,\mathrm{d}t. \end{aligned}$$

故 $F(p)$ 在 $\mathrm{Re}\,p > \sigma_c$ 内解析。
Hence $F(p)$ is analytic in $\mathrm{Re}\,p > \sigma_c$.

Notes:

原稿补充说明
Additional Remarks in the Original Notes

1. “由上述证明中知，$F(p)$ 解析且 $F(+\infty) = 0$。”
   “The preceding proof shows that $F(p)$ is analytic and $F(+\infty) = 0$.”
2. “由解析延拓，$F(p)$ 于 p 平面上除若干孤立奇点外都解析。”
   “By analytic continuation, $F(p)$ is analytic throughout the p-plane except at certain isolated singularities.”

Notes:

3. 常用拉氏变换对
Common Laplace Transform Pairs

教材 P.345，共 73 对”。
textbook, p. 345.“73 pairs in total”.

第 1 对：单位阶跃函数
Pair 1: The Unit Step Function

$$u(t) \overset{\mathcal{L}}{\leftrightarrow} \frac{1}{p}, \qquad \mathrm{Re}\,p > 0.$$

因为
Since

$$\begin{aligned}\mathcal{L}[u(t)] &= \int_0^{+\infty} 1 \cdot e^{-pt}\,\mathrm{d}t \\ &= \left.\frac{e^{-pt}}{-p}\right|_0^{+\infty} \\ &= \frac{1}{p}.\end{aligned}$$

第 2 对：幂函数
Pair 2: Power Functions

类似地，
Similarly,

$$\begin{aligned}\mathcal{L}[t] &= \int_0^{+\infty} te^{-pt}\,\mathrm{d}t \\ &= \int_0^{+\infty} t\,\mathrm{d}\left(\frac{e^{-pt}}{-p}\right) \\ &= \left.-\frac{te^{-pt}}{p}\right|_0^{+\infty} + \int_0^{+\infty} \frac{e^{-pt}}{p}\,\mathrm{d}t \\ &= 0 + \left.\frac{e^{-pt}}{-p^2}\right|_0^{+\infty} \\ &= \frac{1}{p^2}.\end{aligned}$$

$$\begin{aligned}\mathcal{L}[t^2] &= \int_0^{+\infty} t^2e^{-pt}\,\mathrm{d}t \\ &= \left.t^2\frac{e^{-pt}}{-p}\right|_0^{+\infty} - \int_0^{+\infty} \frac{e^{-pt}}{-p}\,2t\,\mathrm{d}t \\ &= \frac{2}{p}\cdot\frac{1}{p^2} \\ &= \frac{2!}{p^3}.\end{aligned}$$

推广为
Generalizing gives

$$\mathcal{L}[t^n] = \frac{n!}{p^{n+1}},$$

即
That is,

$$t^n \overset{\mathcal{L}}{\leftrightarrow} \frac{n!}{p^{n+1}}, \qquad n = 0, 1, 2, \ldots, \quad \mathrm{Re}\,p > 0.$$

第 4 对：指数函数

Pair 4: An Exponential Function

$$e^{at} \overset{\mathcal{L}}{\leftrightarrow} \frac{1}{p-a},$$

其中 a 为常数。

where a is a constant.

$$\begin{aligned}\mathcal{L}[e^{at}] &= \int_0^{+\infty} e^{at}e^{-pt}\,\mathrm{d}t \\ &= \left.\frac{e^{(a-p)t}}{a-p}\right|_0^{+\infty} \\ &= \frac{1}{p-a},\end{aligned}$$

收敛条件为

The convergence condition is

$$\mathrm{Re}(a-p) < 0 \quad \Longleftrightarrow \quad \mathrm{Re}\,p > \mathrm{Re}\,a.$$

第 20—23 对：三角函数与双曲函数

Pairs 20–23: Trigonometric and Hyperbolic Functions

原编号 / Original no.	原函数 / Function	拉氏变换 / Laplace transform
20	$\sin\beta t$	$\dfrac{\beta}{p^2+\beta^2}$
21	$\cos\beta t$	$\dfrac{p}{p^2+\beta^2}$
22	$\sinh\beta t$	$\dfrac{\beta}{p^2-\beta^2}$
23	$\cosh\beta t$	$\dfrac{p}{p^2-\beta^2}$

注：当 β 为实数时，前两式可取 $\mathrm{Re}\,p > 0$，后两式可取 $\mathrm{Re}\,p > |\beta|$。

Editorial note. For real β, the first two formulas hold for $\mathrm{Re}\,p > 0$, and the last two hold for $\mathrm{Re}\,p > |\beta|$.

以正弦函数为例

Derivation for the sine function

$$\begin{aligned}\mathcal{L}[\sin\beta t] &= \mathcal{L}\left[\frac{e^{i\beta t} - e^{-i\beta t}}{2i}\right] \\ &= \frac{1}{2i}\left\{\mathcal{L}[e^{i\beta t}] - \mathcal{L}[e^{-i\beta t}]\right\} \\ &= \frac{1}{2i}\left[\frac{1}{p-i\beta} - \frac{1}{p+i\beta}\right] \\ &= \frac{1}{2i}\cdot\frac{2i\beta}{p^2+\beta^2} \\ &= \frac{\beta}{p^2+\beta^2}.\end{aligned}$$

Notes:

四、拉氏变换的性质

Properties of the Laplace Transform

教材 *P.214*。

textbook, p. 214.

1. 性质 1：线性性

Property 1: Linearity

$$\mathcal{L}[af_1 + bf_2] = a\mathcal{L}[f_1] + b\mathcal{L}[f_2].$$

$$\mathcal{L}^{-1}[aF_1 + bF_2] = a\mathcal{L}^{-1}[F_1] + b\mathcal{L}^{-1}[F_2].$$

例：求拉氏反变换

Example: Find the Inverse Laplace Transform

$$\mathcal{L}^{-1}\left[\frac{p}{(p-1)(p^2+4)}\right].$$

先作部分分式分解：

First, decompose the expression into partial fractions.

$$\begin{aligned}\frac{p}{(p-1)(p^2+4)} &= \frac{A}{p-1} + \frac{Bp+C}{p^2+4} \\ &= \frac{1/5}{p-1} + \frac{-p/5+4/5}{p^2+4}.\end{aligned}$$

所以

Therefore,

$$\begin{aligned}\mathcal{L}^{-1}\left[\frac{p}{(p-1)(p^2+4)}\right] &= \frac{1}{5}\mathcal{L}^{-1}\left[\frac{1}{p-1}\right] - \frac{1}{5}\mathcal{L}^{-1}\left[\frac{p}{p^2+4}\right] \\ &\quad + \frac{4}{5}\mathcal{L}^{-1}\left[\frac{1}{p^2+4}\right] \\ &= \frac{1}{5}e^t - \frac{1}{5}\cos 2t + \frac{2}{5}\sin 2t.\end{aligned}$$

原稿红字提示所用变换对：

The original notes highlight the following transform pairs in red.

$$e^{at} \overset{\mathcal{L}}{\leftrightarrow} \frac{1}{p-a},$$

$$\cos at \overset{\mathcal{L}}{\leftrightarrow} \frac{p}{p^2+a^2},$$

$$\sin at \overset{\mathcal{L}}{\leftrightarrow} \frac{a}{p^2+a^2}.$$

Notes:

2. 性质 2：相似性

Property 2: Scaling

设

Let

$$f(t) \overset{\mathcal{L}}{\leftrightarrow} F(p).$$

则

then

$$f(at) \overset{\mathcal{L}}{\leftrightarrow} \frac{1}{a}F\left(\frac{p}{a}\right), \qquad a>0.$$

以及

and

$$\frac{1}{a}f\left(\frac{t}{a}\right) \overset{\mathcal{L}}{\leftrightarrow} F(ap).$$

Notes:

3. 性质 3：延迟性

Property 3: Time Delay

“对 $f(t)u(t)$ 的平移性”。

“shifting $f(t)u(t)$”.

$$f(t-t_0)u(t-t_0) \overset{\mathcal{L}}{\leftrightarrow} e^{-pt_0}F(p).$$

即

That is,

$$\mathcal{L}[f(t-t_0)u(t-t_0)] = e^{-pt_0}F(p),$$

以及
and

$$\mathcal{L}^{-1}[e^{-pt_0}F(p)] = f(t-t_0)u(t-t_0).$$

注：单边拉普拉斯变换的这一延迟公式通常取 $t_0 \geq 0$。
note. For the unilateral Laplace transform, this time-delay formula is normally used with $t_0 \geq 0$.

Notes:

4. 性质 4：对 $F(p)$ 的平移性

Property 4: Shifting in the Transform Variable

$$e^{p_0 t}f(t) \overset{\mathcal{L}}{\leftrightarrow} F(p-p_0).$$

原稿证明从定义开始：
The original proof starts from the definition.

$$F(p) = \int_0^{+\infty} f(t)e^{-pt}\,\mathrm{d}t.$$

本页到此未继续展开证明。
Notes:

五、12.1 作业

Assignment for December 1

- 教材 P.204，A 组第 9、10 题。
  Textbook, p. 204: Group A, Exercises 9 and 10.
- 教材 P.204，B 组第 1、2、5、6、8 题。
  Textbook, p. 204: Group B, Exercises 1, 2, 5, 6, and 8.
- 教材 P.236，第 1 题。
  Textbook, p. 236: Exercise 1.

Notes: